

Lecture 17: Bernoulli Process

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$$P(T = k) = (1 - p)^{k-1}p, \quad k = 1, 2, \dots \quad (\text{Geometric}(p)), \quad E[T] = \frac{1}{p}.$$

- :::—::: {.callout-warning icon=false}## FactMemorylessness: $P(T > s + t \mid T > s) = P(T > t)$ for integers $s, t \geq 0$ – waiting doesn't help or hurt.:...## Simulating a Bernoulli process 1
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—title: “Lecture 17: Bernoulli Process”author: “Jamie Haddock”format: revealjs: output-file: Lecture17_slides slide-number: true preview-links: auto logo: figs/hmc.png css: input/slides.css incremental: true smaller: true code-fold: true embed-resources: true html: code-fold: true embed-resources: true pdf: documentclass: article toc: true number-sections: true geometry: - top=1in - left=1in - bottom=1in - right=1informat-links: falsejupyter: python3filters: - input/remove-pause.luaexecute: echo: true eval: true—

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```
from scipy.stats import geom p = 0.15 trials = rng.random(4000) < p success_times = np.where(trials)[0] intervals = np.diff(success_times) s, t = 3, 5 print(f"P(T > s+t | T > s) ~ {np.mean(intervals[intervals > s] - s > t):.3f}") print()
```

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